

Density of materials

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 4 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	Define density.	The mass per unit volume of a substance.	☐
2	State the standard unit of volume.	Cubic metre (m ³)	☐
3 ★	State the standard unit of density.	Kilogram per cubic metre (kg/m ³)	☐
4	Which state of matter can be compressed easily?	Gas	☐
5	Explain why solids and liquids cannot be compressed easily.	Their particles are already touching.	☐
6 ★	Describe the motion of particles in a solid.	They vibrate around a fixed position.	☐
7	Describe the motion of particles in a liquid.	They move randomly, sliding past one another.	☐
8	Describe the motion of particles in a gas.	They move quickly in random directions.	☐
9	State the equation linking density, mass and volume.	density = mass ÷ volume ($\rho = m \div V$)	☐
10	Describe the arrangement of the particles in a solid.	Very close together in a regular pattern.	☐
11	Describe the arrangement of the particles in a liquid.	Close together, in a random arrangement.	☐
12	Describe the arrangement of the particles in a gas.	Far apart, in a random arrangement.	☐
13 ★	Explain why gases have a much lower density than solids and liquids.	The particles are far apart, so there is less mass in a given volume.	☐